Engineering Challenges

Section Européenne – Sciences de l'Ingénieur

EC01 to EC30

Contents

EC01 – Renewable Energy

Can solar power meet tomorrow's energy needs?

Learning Objectives

- Explain how a photovoltaic cell and a grid-connected PV system operate.
- Interpret capacity data and calculate annual additions.
- Discuss intermittency, storage, grid integration and recycling.

Engineering News

Global renewable capacity grew by a record 692 GW in 2025. Solar power alone accounted for about 510 GW of additions, nearly three-quarters of the renewable total. The engineering challenge is now not only to manufacture more panels, but also to connect variable generation to stronger and more flexible grids.

Main Article

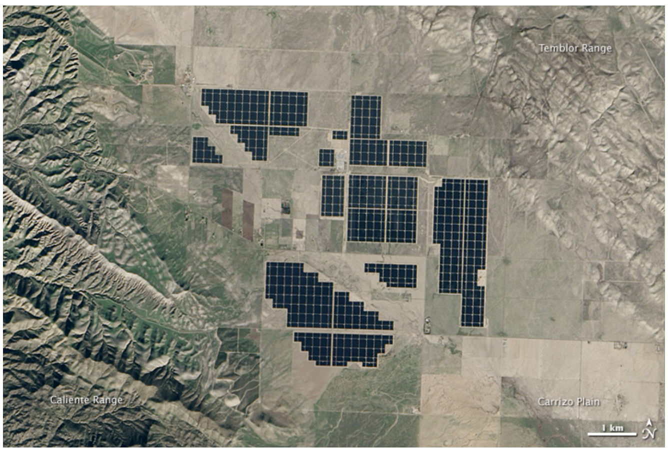
A photovoltaic (PV) cell converts light directly into electricity. Most commercial cells are manufactured from silicon, a semiconductor. When photons provide enough energy, electrons can move through the material. The internal electric field of the p–n junction separates the electrical charges and creates a direct current.

A PV module connects many cells together. Modules are combined into strings and arrays to reach the required voltage and power. However, a module rarely operates at a fixed point. Its voltage-current characteristic changes with solar irradiance and temperature. A maximum power point tracker continuously adjusts the operating point so that the array delivers as much power as possible.

The generated electricity is direct current, whereas buildings and public grids use alternating current. An inverter therefore converts DC into AC, synchronises the voltage and frequency with the grid, and provides protection and monitoring functions. A battery may store surplus electricity, but it increases cost, material use and conversion losses.

Solar power is variable: output changes with clouds, season and time of day. Engineers integrate large amounts of PV using forecasting, flexible demand, interconnections, storage and controllable generation. The objective is not to make every installation independent, but to balance the complete electricity system.

The environmental impact of a PV system must be assessed over its entire life cycle. Manufacturing requires energy and materials, while end-of-life modules contain glass, aluminium, copper, polymers and semiconductor materials. Designing modules that are efficient, durable, repairable and recyclable is therefore a major engineering goal.



Topaz Solar Farm, California. NASA image, public domain.

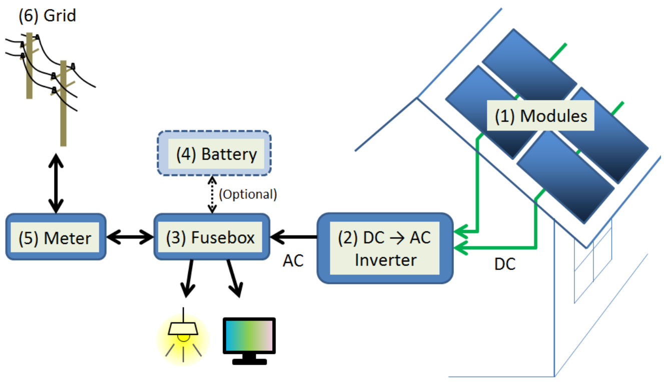
Engineering Focus

Designing a photovoltaic installation

Engineers must optimise several parameters simultaneously:

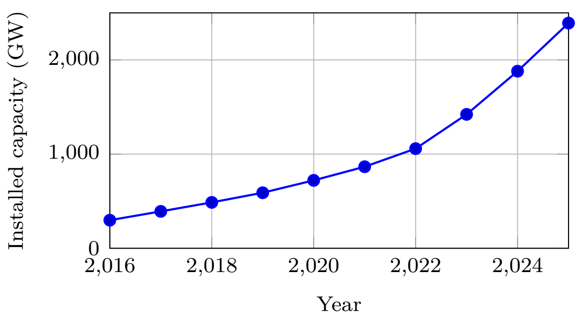
- panel orientation and tilt;
- inverter sizing and MPPT architecture;
- cable cross-sections and electrical losses;
- thermal behaviour and ventilation;
- protection and isolation devices;
- monitoring and maintenance strategy;
- expected lifetime and recyclability.

A good design minimises the levelised cost of electricity while ensuring safety, reliability and high annual energy yield.



Simplified residential PV system. S-kei, CC0.

Global solar PV capacity



Source: IRENA, Renewable Capacity Statistics 2026.

Did You Know?

The solar energy reaching the Earth in roughly one hour is comparable to humanity's annual energy use. The main engineering challenge is therefore to capture, convert, store and distribute only a small fraction of this resource efficiently.

Key Figures

Commercial module efficiency	20–24%
Typical module lifetime	25–30 years
Typical inverter efficiency	97–99%
Annual module degradation	0.3–0.5%/year
Global PV capacity in 2025	$\approx$ 2.4 TW

Useful Vocabulary

solar cell	cellule photovoltaïque
irradiance	irradiance
inverter	onduleur
grid	réseau électrique
storage	stockage
yield	production énergétique
lifetime	durée de vie
recycling	recyclage

Questions

1. Explain the role of the p–n junction.
2. Why does a PV installation need an inverter?
3. What is the purpose of MPPT control?
4. Use the graph to estimate the increase between 2020 and 2025.
5. Give three technical solutions for integrating variable solar power.
6. Why must environmental assessment include manufacturing and end-of-life?

Engineering Challenge

Your school consumes 410 MWh of electricity each year and has 850 m^2 of usable roof area. Prepare a preliminary PV study:

1. estimate the peak power that could be installed;
2. estimate annual energy production;
3. discuss the likely self-consumption ratio;
4. decide whether battery storage is justified;
5. propose useful monitoring indicators.

Present your conclusions as a short engineering report.

Speaking Task

Working in pairs, prepare a five-minute presentation answering the question: "Can Europe rely mainly on solar electricity by 2050?" Support your arguments with technical evidence, environmental impacts, economic considerations and examples from different European countries.

EC02 – Hydrogen

Can hydrogen decarbonise heavy transport?

Learning Objectives

- Distinguish grey, blue and green hydrogen.
- Explain the operating principles of electrolysers and fuel cells.
- Compare hydrogen and battery-electric energy chains.

Engineering News

Low-emission hydrogen projects are being developed for fertiliser production, steel, shipping and heavy road transport. Their success depends not only on electrolyser capacity, but also on electricity supply, storage, transport and final demand.

Main Article

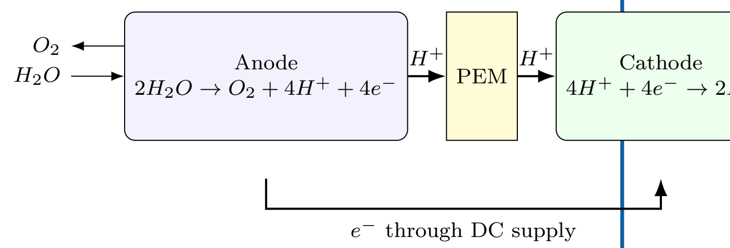
Hydrogen is an **energy carrier**, not a primary source of energy. It must be manufactured from another resource before it can be stored and used. Most hydrogen is still produced from natural gas by steam methane reforming. If the resulting carbon dioxide is released, the product is called grey hydrogen. Capturing part of the CO_2 leads to so-called blue hydrogen. Green hydrogen is produced by electrolysis powered by low-carbon renewable electricity.

In a proton-exchange-membrane (PEM) electrolyser, water enters the anode side. An electric current separates the water molecules into oxygen, protons and electrons. The membrane allows protons to cross while forcing electrons through an external circuit. Hydrogen is formed at the cathode.

A PEM fuel cell performs the reverse overall reaction. Hydrogen is supplied to the anode and oxygen from air reaches the cathode. The electrochemical reaction produces electricity, water and heat without combustion. Several cells are assembled in a stack to obtain the required voltage and power.

Hydrogen has a high specific energy by mass and enables rapid refuelling. However, its low volumetric density requires compression, liquefaction or conversion into another carrier. Each conversion consumes energy. Consequently, a battery-electric vehicle is usually more efficient from electricity source to wheels, while hydrogen can remain attractive when long range, high payload and short refuelling stops are essential.

PEM electrolyser



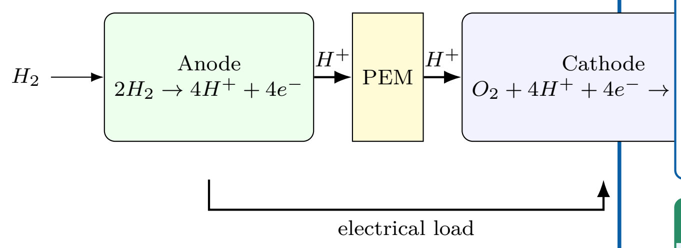
Engineering Focus

The complete hydrogen value chain

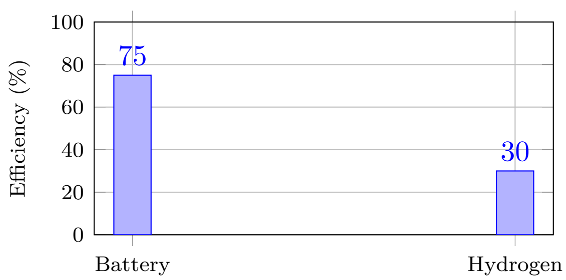
1. Generate low-carbon electricity.
2. Produce hydrogen by electrolysis.
3. Purify and compress the gas.
4. Store and distribute it safely.
5. Convert it into useful energy in a fuel cell or industrial process.

Engineers optimise efficiency, safety, materials, cost and infrastructure at every stage. Hydrogen leaks must also be controlled because the molecule is very small and has a wide flammability range in air.

PEM fuel cell



Electricity-to-wheel efficiency



Illustrative whole-chain values; actual performance depends on equipment and operating conditions.

Did You Know?

Hydrogen contains about 120 MJ of energy per kilogram on a lower-heating-value basis, but storing one kilogram requires a much larger tank than storing an equivalent amount of liquid fuel.

Key Figures

Hydrogen LHV	≈ 120 MJ/kg
Electrolyser efficiency	60–75%
Fuel-cell electrical efficiency	40–60%
Typical vehicle storage pressure	350 or 700 bar

Useful Vocabulary

electrolyser	électrolyseur
fuel cell	pile à combustible
stack	empilement de cellules
compressed gas	gaz comprimé
refuelling	ravitaillement
leak detection	détection de fuite

Questions

1. Why is hydrogen described as an energy carrier?
2. Compare grey, blue and green hydrogen.
3. Explain the role of the PEM in an electrolyser.
4. What products leave a fuel cell?
5. Why is the battery chain usually more efficient?
6. In which transport applications can hydrogen remain attractive?

Engineering Challenge

A logistics company operates 120 long-haul trucks. Compare battery-electric and hydrogen fuel-cell solutions using efficiency, range, payload, refuelling time, infrastructure and life-cycle emissions. Present a justified engineering recommendation.

Speaking Task

Debate: "Hydrogen will replace batteries for heavy transport." Each team must use at least three technical arguments and respond to one opposing argument.

Sources and further reading

IEA — Global Hydrogen Review: online resource

;

U.S. Department of Energy — Hydrogen and Fuel Cell Technologies Office: online resource

;

European Commission — Hydrogen: online resource

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EC03 – Wind Energy

How do engineers harvest the wind?

Learning Objectives

- Explain how aerodynamic lift produces rotor torque.
- Identify the main subsystems of a modern wind turbine.
- Interpret a wind-turbine power curve.
- Compare onshore, fixed-bottom offshore and floating offshore solutions.

Engineering News

Modern offshore wind turbines now reach ratings above 15 MW. Floating foundations make it possible to exploit deep-water sites where winds are stronger and more regular, but they also introduce new challenges in mooring, maintenance and grid connection.

Key Figures

Rotor diameter	120–250 m
Rated power	3–18 MW
Cut-in speed	3–4 m/s
Rated speed	11–15 m/s
Cut-out speed	20–25 m/s
Capacity factor	30–60%

Main Article

A wind turbine converts the kinetic energy of moving air into electrical energy. The blades are shaped like aerofoils. As air flows around a blade, a pressure difference creates lift. The tangential component of this force produces a torque on the rotor.

The available power in the wind is

$$P_{\text{wind}} = \frac{1}{2}\rho A v^3,$$

where ρ is air density, A the swept area and v wind speed. The cubic term explains why site selection is critical: a small increase in average wind speed can produce a large increase in annual energy output.

The rotor drives either a gearbox and a high-speed generator or a direct-drive generator. Power electronics convert the variable-frequency output into electricity compatible with the grid. Pitch control adjusts the blade angle, while yaw control rotates the nacelle so that the rotor faces the wind.

Engineering Focus

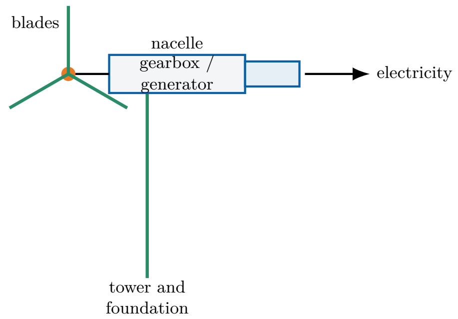
Main subsystems

1. Rotor, blades and hub;
2. Main shaft and bearings;
3. Gearbox or direct-drive generator;
4. Converter and transformer;
5. Pitch and yaw actuators;
6. Tower, foundations and monitoring system.

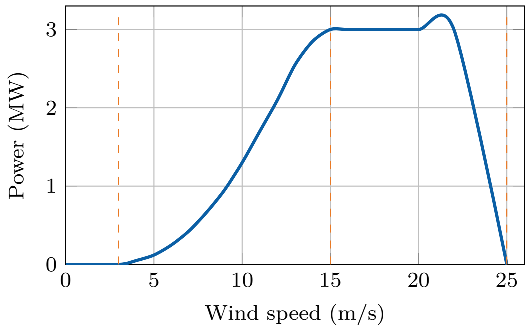
Did You Know?

No turbine can capture all the power in the wind. The theoretical maximum is the **Betz limit**: 59.3% of the kinetic power crossing the rotor area.

Wind-turbine architecture



Typical power curve



Cut-in, rated and cut-out regions determine how the turbine operates safely.

Useful Vocabulary

blade	pale
hub	moyeu
gearbox	multiplicateur
yaw	orientation de la nacelle
pitch	calage des pales
swept area	surface balayée
capacity factor	facteur de charge

Questions

1. Why does wind power depend on the cube of wind speed?
2. What is the purpose of pitch control?
3. What is the purpose of yaw control?
4. Interpret the three regions of the power curve.
5. Compare onshore and offshore wind farms.

Sources

Global Wind Energy Council: online resource

IEA Wind: online resource

Engineering Challenge

A coastal region plans a 500 MW wind farm. Compare an onshore solution, a fixed-bottom offshore solution and a floating offshore solution. Consider energy yield, foundations, maintenance, grid connection, environmental impact and public acceptance. Present a justified recommendation.

Speaking Task

Debate the statement: “Large wind farms should be built mainly offshore.” Use technical, economic and environmental arguments.

EC04 – Nuclear Energy

Can nuclear power support a low-carbon future?

Learning Objectives

- Explain nuclear fission and the controlled chain reaction.
- Identify the energy and fluid flows in a pressurised water reactor (PWR).
- Interpret lifecycle-emission data and discuss the role of nuclear power.

Engineering News

Several countries are studying new reactor programmes, lifetime extensions and Small Modular Reactors (SMRs). The central engineering questions concern construction time, financing, safety demonstration, fuel supply, waste management and integration with an electricity system containing more variable renewable generation.

Main Article

A nuclear power station converts the energy released by **nuclear fission** into electricity. When a uranium-235 nucleus absorbs a neutron, it can split into two lighter nuclei, release energy and emit additional neutrons. These neutrons may trigger further fissions. In a reactor, the chain reaction is kept stable rather than allowed to grow uncontrolled.

In a pressurised water reactor, fuel assemblies are placed inside the reactor vessel. Water in the primary circuit removes heat from the core. It is maintained at high pressure so that it remains liquid at a temperature of roughly 300 °C. In the steam generator, this primary water transfers heat to a separate secondary circuit. The secondary water boils and produces steam, which drives a turbine connected to an electrical generator.

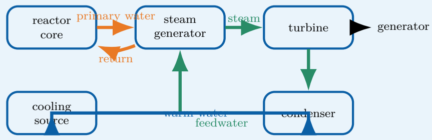
After the turbine, a condenser transforms the steam back into liquid water. A third cooling circuit carries waste heat towards a river, the sea or a cooling tower. The three circuits have different functions and are kept physically separate.

Reactor power is controlled by adjusting neutron absorption. Control rods contain materials that absorb neutrons, while soluble boron may also be used in the primary water. Even after the chain reaction stops, radioactive decay continues to release **residual heat**; cooling must therefore remain available.

Safety relies on several independent barriers and systems: the fuel matrix, the metal cladding, the primary circuit boundary and the containment building. Engineers also analyse extreme events, redundancy, component ageing, human factors, cybersecurity and long-term waste management.

Engineering Focus

Energy conversion in a PWR



The primary circuit transports nuclear heat, the secondary circuit produces steam and the cooling circuit rejects unused heat. The reactor is therefore both a nuclear system and a large thermal power plant.

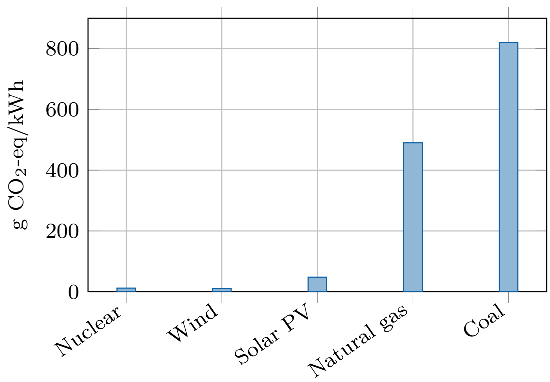
Did You Know?

Stopping fission does not immediately stop heat production. Residual heat falls rapidly, but cooling remains essential for hours, days and longer after shutdown.

Key Figures

Typical French PWR output	900–1450 MW
Primary-water temperature	about 300 °C
Primary pressure	about 155 bar
Typical capacity factor	70–90%
Design operating life	several decades

Indicative lifecycle emissions



Indicative median lifecycle values based on IPCC assessments. Values vary with technology, location and methodology.

Fission and fusion

Fission splits heavy nuclei and is used in present commercial reactors. **Fusion** combines light nuclei and powers the Sun. Fusion research aims to produce controlled net energy, but it is not yet a commercial electricity source.

Useful Vocabulary

chain reaction	réaction en chaîne
reactor vessel	cuve du réacteur
fuel assembly	assemblage combustible
control rod	barre de contrôle
steam generator	générateur de vapeur
containment	enceinte de confinement
residual heat	puissance résiduelle
spent fuel	combustible usé

Questions

1. What happens during nuclear fission?
2. Why must the primary water remain under high pressure?
3. Explain why the PWR contains several separate water circuits.
4. What is the role of control rods?
5. Why is cooling required after reactor shutdown?
6. Use the graph to compare nuclear power with fossil generation.
7. Give two advantages and two limitations of SMRs.

Sources and further reading

IAEA - Nuclear Power Reactors: online resource

OECD Nuclear Energy Agency: online resource

IPCC - Energy Systems: online resource

Engineering Challenge

A region must replace 8 GW of fossil-fuelled generation while reducing lifecycle CO_2 emissions and maintaining electricity supply during winter evenings. Propose a balanced mix using nuclear, wind, solar, storage, interconnections and demand response. Justify:

1. installed capacity;
2. expected availability;
3. flexibility and reserve requirements;
4. environmental and social constraints;
5. construction sequence and major risks.

Present the result as a two-page engineering recommendation.

Speaking Task

Debate: "Europe should build new nuclear power stations." One team supports the proposal and one team opposes it. Use evidence about safety, climate, cost, waste, construction time and security of supply.

EC05 – Energy Storage

Why storing energy is key to the energy transition

Learning Objectives

- Explain why electrical grids need energy storage.
- Compare storage technologies using power, energy, duration and efficiency.
- Select a suitable storage mix for a given engineering problem.

Engineering News

As variable solar and wind generation expands, grid operators are deploying batteries, pumped-storage plants and other forms of flexibility. The key challenge is not to find one universal storage device, but to combine technologies operating over seconds, hours, days and seasons.

Main Article

Electricity networks must keep production and consumption balanced at every instant. A sudden mismatch changes grid frequency and can damage equipment or trigger disconnections. Storage systems absorb energy when production exceeds demand and return it when the system needs additional power.

Engineers distinguish **power** from **energy**. Power, measured in watts, describes how quickly a storage system can charge or discharge. Energy, measured in watt-hours or joules, describes how long it can maintain that power. A flywheel may deliver high power for a few seconds, whereas a reservoir or hydrogen cavern can store energy for much longer periods.

Lithium-ion batteries provide rapid response and high round-trip efficiency. They are well suited to frequency control, electric vehicles and daily solar shifting. Their limitations include cost, ageing, fire protection and the supply of raw materials. Pumped-storage hydroelectricity stores much larger amounts of energy: water is pumped to an upper reservoir and later released through a reversible turbine. Its lifetime is long, but suitable terrain and major civil works are required.

Compressed air, thermal storage and hydrogen address longer durations. Hydrogen is produced by electrolysis, stored, and later used in a fuel cell or turbine. Because each conversion loses energy, the round-trip efficiency is lower than that of a battery. Nevertheless, hydrogen may become useful when storage duration matters more than efficiency, particularly for seasonal balancing or industrial fuel supply.

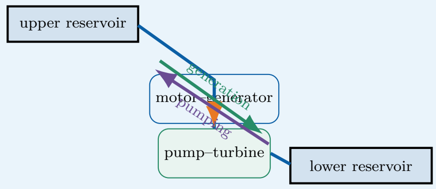
No storage option is optimal for every task. A resilient low-carbon grid therefore combines storage with interconnections, demand response, controllable generation and accurate forecasting.

Did You Know?

A storage system can have excellent efficiency but still be unsuitable if its discharge duration, response time, lifetime or location does not match the service required by the grid.

Engineering Focus

Pumped-storage hydroelectricity

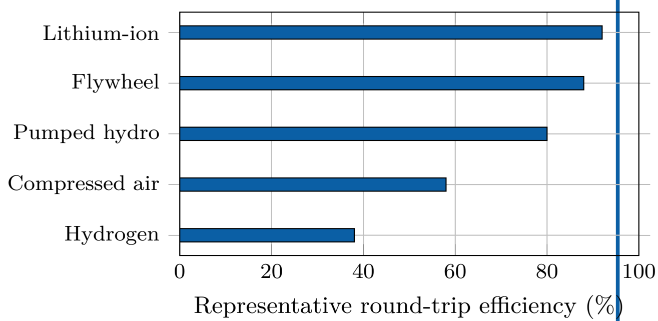


During low demand, electrical energy drives the pump and raises water. During high demand, the water flows downhill and drives the turbine. The stored gravitational energy is $E = mgh$.

Key Figures

Lithium-ion round-trip efficiency	85–95%
Pumped hydro	70–85%
Hydrogen power-to-power	30–45%
Typical battery duration	1–8 h
Seasonal storage	weeks–months

Storage technologies compared



Representative values only: actual performance depends on design and operating conditions.

Useful Vocabulary

round-trip efficiency	rendement aller-retour
charge / discharge	charge / décharge
storage duration	durée de stockage
reservoir	réservoir
flywheel	volant d'inertie
demand response	effacement de consommation
grid frequency	fréquence du réseau

Questions

1. Why must electricity production and consumption remain balanced?
2. Distinguish power from stored energy.
3. Why are batteries suitable for rapid grid services?
4. Explain the operating cycle of a pumped-storage plant.
5. Why can hydrogen be useful despite its lower efficiency?
6. Which criteria should engineers consider besides efficiency?

Engineering Challenge

A remote island has a 12 MW peak demand, a 20 MW wind farm and a 10 MW solar farm. Propose a storage strategy for frequency control, night-time supply and a five-day period of weak wind. Select at least two technologies and justify their power, capacity, duration, efficiency, safety and environmental impact.

Speaking Task

In teams, defend one storage technology in a three-minute technical pitch. Then identify one service for which your technology would be a poor choice.

Sources

IEA – Grid-scale storage: online resource

NREL – Energy storage: online resource

EC06 – Electric Vehicles

Are electric cars really sustainable?

Learning Objectives

- Describe the main components of a battery-electric vehicle powertrain.
- Explain charging, regenerative braking and battery thermal management.
- Assess an electric-vehicle project using technical and environmental criteria.

Engineering News

Electric mobility is moving beyond the vehicle itself. Engineers now design complete systems that connect cars, charging stations, buildings and electricity networks. Smart charging can shift demand away from peak hours, while vehicle-to-grid operation may allow parked vehicles to provide flexibility to the power system.

Did You Know?

An electric motor can reverse its energy flow in a fraction of a second: it drives the wheels during acceleration and becomes a generator during regenerative braking.

Main Article

A battery-electric vehicle stores electrical energy in a high-voltage battery pack. During acceleration, the battery supplies direct current to a power-electronics inverter. The inverter controls the alternating current delivered to the traction motor, which converts electrical energy into mechanical torque at the wheels. Unlike a combustion engine, an electric motor can produce high torque from very low speed and commonly operates with high efficiency over a broad range of conditions.

The battery is not simply a collection of cells. It also includes electrical connections, contactors, sensors, cooling channels, structural protection and a **Battery Management System** (BMS). The BMS estimates state of charge and state of health, balances the cells and protects the pack against excessive current, voltage and temperature. Thermal management is critical because low temperatures reduce available power, while high temperatures accelerate ageing and may increase safety risks.

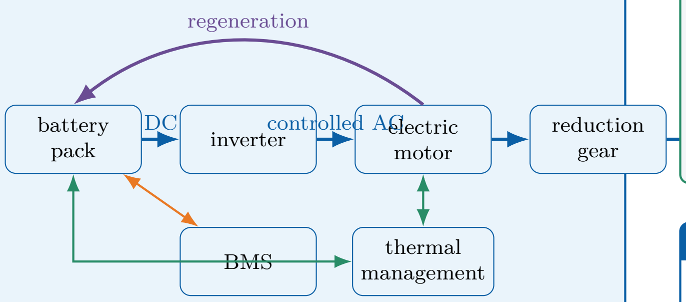
When the driver brakes, the inverter can operate the traction motor as a generator. Part of the vehicle's kinetic energy is then converted back into electrical energy and stored in the battery. This **regenerative braking** reduces energy consumption and mechanical-brake wear, although friction brakes remain necessary for emergency stops, low-speed operation and situations in which the battery cannot accept additional power.

Charging power depends on both the charging station and the vehicle. An on-board charger converts alternating current from a domestic or public AC supply. Fast DC charging bypasses the on-board charger and feeds controlled direct current to the battery. High charging power shortens stops, but it also increases heat generation, infrastructure cost and stress on the local electricity network.

Electric vehicles have no exhaust emissions during use, but they are not impact-free. Engineers must consider battery manufacture, electricity generation, vehicle mass, tyre particles, lifetime and recycling. Their environmental performance therefore depends on the complete life cycle and on how the electricity used for charging is produced.

Engineering Focus

Energy flow in a battery-electric vehicle

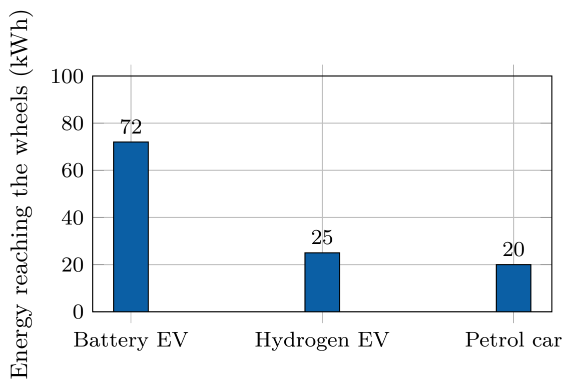


The high-voltage traction circuit is complemented by a DC/DC converter that supplies the vehicle's low-voltage network. Isolation monitoring and contactors disconnect the battery if a serious fault is detected.

Key Figures

Electric motor peak efficiency	90–97%
Typical battery voltage	350–800 V
Home AC charging	2–11 kW
Rapid DC charging	50–350 kW
Typical passenger-car battery	40–100 kWh

From supplied energy to the wheels



Illustrative conversion chain starting from 100 kWh of supplied energy. Actual values vary with technology, driving conditions and system boundaries.

Useful Vocabulary

battery pack	pack batterie
state of charge	état de charge
state of health	état de santé
traction motor	moteur de traction
regenerative braking	freinage régénératif
on-board charger	chargeur embarqué
charging point	point de recharge
power electronics	électronique de puissance

Questions

1. What are the functions of the inverter and the traction motor?
2. Why does a battery pack require thermal management?
3. Explain how regenerative braking changes the direction of energy flow.
4. Compare AC charging with rapid DC charging.
5. Why can charging power be limited by the vehicle rather than the station?
6. Which life-cycle impacts must be considered beyond exhaust emissions?
7. Why is smart charging useful for the electricity network?

Engineering Challenge

A city operates 24 buses, each travelling 220 km per day. Compare overnight depot charging with high-power opportunity charging at selected terminal stops. Propose a battery capacity, charging power and operating schedule. Justify your choices using range, charging time, battery ageing, network connection, redundancy, safety and maintenance.

Speaking Task

Prepare a four-minute technical briefing answering the question: "Should all new urban cars be electric by 2035?" Present one technical advantage, one limitation and one policy measure that could improve the overall transport system.

Sources

IEA – Global EV Outlook: online resource

U.S. Department of Energy – Electric vehicles: online resource

Alternative Fuels Data Center – How electric vehicles work: online resource

EC07 – Robotics

How can robots work safely with humans?

Learning Objectives

- Identify the mechanical, electrical and software subsystems of a robot.
- Explain position, force and vision feedback in a collaborative workstation.
- Propose a safe robotic solution for an industrial task.

Engineering News

Collaborative robots, or **cobots**, are changing the organisation of factories and laboratories. Instead of operating permanently behind a fence, they can share selected tasks with people. The engineering objective is not merely to make the robot slower: the complete workstation must detect hazards, limit energy and bring the system to a safe state when required.

Did You Know?

A collaborative robot is not automatically a safe collaborative application. The gripper, the workpiece, the programmed speed and the surrounding equipment must all be included in the risk assessment.

Main Article

A robot is a programmable machine that senses its environment, processes information and acts through a mechanical structure. A typical articulated robot contains rigid links connected by joints. Electric servomotors create torque at these joints, while encoders measure angular position and speed. A controller compares the measured motion with the requested trajectory and continuously corrects the motor commands.

The tool attached to the final link is called the **end-effector**. It may be a gripper, a welding torch, a screwdriver or a camera. Selecting this tool requires engineers to consider payload, accuracy, cycle time, gripping force and the geometry of the product. The useful region that the end-effector can reach is the robot's workspace.

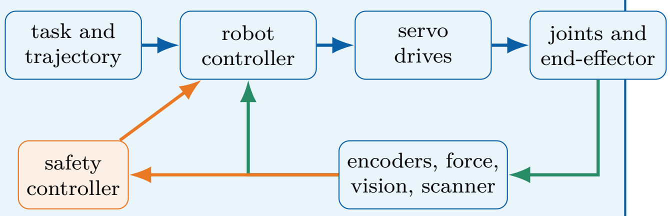
Position control is sufficient for many repetitive tasks, but physical interaction requires additional information. A force–torque sensor can detect contact, while motor-current monitoring can estimate resistance at a joint. Machine-vision systems locate parts, inspect surfaces and track people. These measurements form feedback loops that allow the robot to adapt rather than repeat a completely fixed sequence.

Safe collaboration depends on the entire application. Possible strategies include monitored stops, hand-guiding, speed and separation monitoring, and power-and-force limitation. A scanner or camera can define protective zones around the robot. As a person approaches, the controller first reduces speed and then stops the hazardous motion. Mechanical design also matters: rounded surfaces, low moving mass and the absence of crushing points can reduce injury risk.

Mobile robots combine localisation, mapping and path planning. Because sensors can fail or be obstructed, their design needs diagnostics and a defined safe response. Cybersecurity is equally important: access control, validated software and network segmentation help prevent dangerous commands or corrupted sensor data.

Engineering Focus

Closed-loop architecture of a collaborative robot

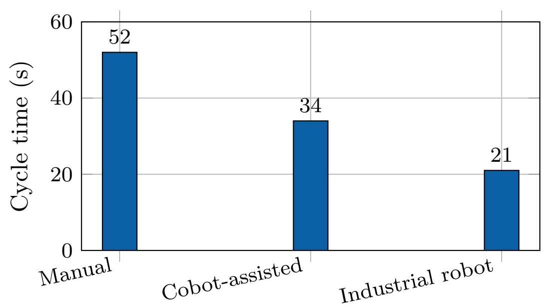


The standard motion controller aims to achieve the task. The safety controller independently supervises hazardous conditions and can request a controlled stop or remove drive power.

Key Figures

Typical articulated cobot axes	6–7
Repeatability of small industrial robots	about 0.02–0.1 mm
Typical cobot payload range	3–35 kg
Main feedback period	milliseconds
Emergency response objective	safe state

Assembly cycle comparison



Illustrative values for one assembly operation. A faster cycle does not by itself prove that a solution is safer, more flexible or more economical.

Useful Vocabulary

joint	articulation
link	solide / bras
actuator	actionneur
encoder	codeur
end-effector	effecteur terminal
payload	charge utile
repeatability	répétabilité
protective stop	arrêt de protection

Questions

1. What information is provided by an encoder?
2. Distinguish accuracy from repeatability.
3. Why may position control be insufficient during contact tasks?
4. Explain speed and separation monitoring.
5. Why must the end-effector be included in the risk assessment?
6. Give two possible consequences of a cybersecurity failure.
7. Why is productivity data alone insufficient when selecting a robot?

Engineering Challenge

A company wants to automate the insertion and tightening of four screws on a 2.5 kg product while an operator performs the adjacent wiring task. Design the collaborative workstation. Select the robot, end-effector and sensors; define the work sequence and protective zones; identify crushing, impact, tool and cybersecurity hazards; and propose validation tests. Compare the solution with a fully manual station and a fenced industrial robot.

Speaking Task

Prepare a four-minute design review answering the question: “Should this workstation use a cobot or a conventional industrial robot?” Defend your choice using safety, cycle time, flexibility, payload, maintenance and cost.

Sources

International Federation of Robotics – industrial and service robots: online resource
NIST – robotics and autonomous systems: online resource
European Commission – machinery safety: online resource

EC08 – Artificial Intelligence

Can machines make reliable engineering decisions?

Learning Objectives

- Distinguish rule-based programming, machine learning and inference.
- Explain how data quality affects classification performance.
- Propose a validated AI architecture for an engineering application.

Engineering News

Artificial intelligence is now embedded in industrial inspection, predictive maintenance, energy management, medical imaging and autonomous systems. Its engineering value comes from its ability to process large quantities of sensor data, but performance figures alone are not enough: reliability, traceability and human responsibility must also be designed.

Did You Know?

An AI model can achieve a high overall accuracy while still missing most rare defects. When the classes are unbalanced, precision and recall are often more informative than accuracy alone.

Main Article

Traditional software follows explicit rules written by a programmer. A machine-learning system instead learns a mathematical model from examples. During **training**, an algorithm adjusts internal parameters so that the model's outputs approach the expected answers contained in a labelled dataset. During **inference**, the trained model processes new data and produces a classification, numerical estimate or recommended action.

In computer vision, a neural network can learn to distinguish acceptable products from defective ones. The input may be an image of a machined surface, while the output is a defect category and a confidence score. In predictive maintenance, vibration, temperature and current measurements can be used to estimate the condition of a bearing or motor. In both cases, the model is only one part of the complete engineering system: sensors, lighting, communication, computing hardware and the human-machine interface also determine the final result.

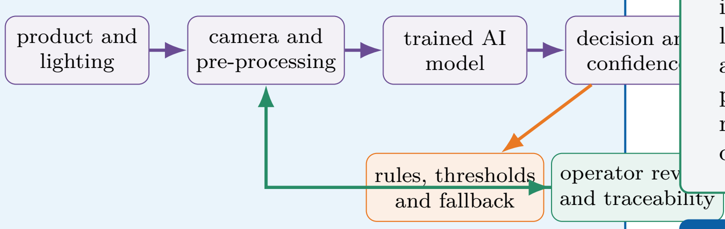
Dataset quality is critical. If defects are rare, poorly labelled or photographed under only one lighting condition, the model may appear accurate while failing in production. Engineers therefore separate data into training, validation and test sets. The test set must remain independent so that it measures generalisation rather than memorisation. Useful indicators include precision, recall, false-alarm rate and missed-detection rate.

Safety-critical applications require additional safeguards. A confidence threshold can prevent uncertain outputs from triggering an automatic decision. Independent physical checks, plausibility rules and human confirmation can also reduce risk. Engineers must monitor **data drift**: over time, new products, sensor ageing or environmental changes may make the operating data different from the original training data.

Explainability and traceability are also important. The development team should record dataset versions, model parameters, test results and software updates. If the system influences a hazardous action, responsibility remains with the organisation that designed and validated the complete system, not with the algorithm itself.

Engineering Focus

Validated AI inspection architecture

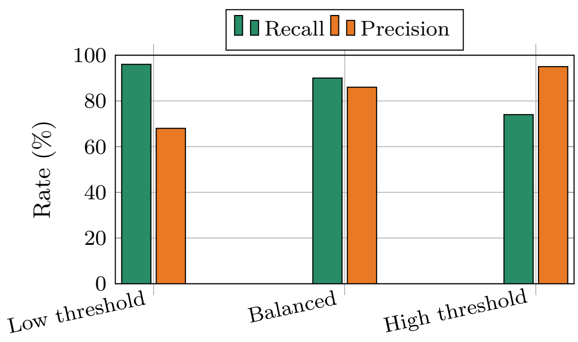


The AI model proposes a result. Independent thresholds and fallback rules determine whether the result may be accepted automatically or must be reviewed by a person. Production data and operator feedback are stored so that performance can be monitored over time.

Key Figures

Training set	model fitting
Validation set	parameter selection
Test set	final independent evaluation
False negative	missed defect
False positive	false alarm
Confidence threshold	application-dependent

Defect-detection trade-off



Illustrative values. Lowering the threshold detects more defects but may also create more false alarms. The correct setting depends on the consequences of each error.

Useful Vocabulary

dataset	jeu de données
training	entraînement
inference	inférence
label	étiquette / classe
accuracy	exactitude globale
precision	précision positive
recall	taux de détection
data drift	dérive des données

Questions

1. Distinguish training from inference.
2. Why must a test set remain independent?
3. Explain the difference between a false positive and a false negative.
4. Why can overall accuracy be misleading for rare defects?
5. What is the purpose of a confidence threshold?
6. Give two possible causes of data drift.
7. Why must responsibility remain with the engineering organisation?

Engineering Challenge

A factory wants to inspect 30 aluminium parts per minute using a camera and an AI model. A missed crack may lead to a dangerous failure, while a false alarm sends a good part for manual inspection. Design the complete inspection station. Specify the camera and lighting arrangement, dataset, performance indicators, acceptance threshold, fallback strategy, traceability records and validation tests. Explain how you would monitor the system after deployment.

Speaking Task

Prepare a four-minute design review answering the question: "Should this inspection system reject parts automatically?" Defend your position using safety, productivity, precision, recall, traceability and human supervision.

Sources

NIST – Artificial Intelligence: online resource
OECD – Artificial Intelligence: online resource
European Commission – Artificial Intelligence: online resource

EC09 – Space Engineering

How do engineers design systems for extreme environments?

Learning Objectives

- Identify the main subsystems of an artificial satellite.
- Explain how engineers manage launch loads, vacuum and temperature variations.
- Propose a coherent architecture for a small Earth-observation mission.

Engineering News

Reusable launch vehicles, commercial constellations and small satellites are changing access to space. Lower launch costs make new missions possible, but every spacecraft must still survive an extreme environment and operate for years without direct maintenance.

Main Article

A spacecraft is designed around its **mission**. An Earth-observation satellite must carry a camera or radar payload, point it accurately towards the ground, store the collected data and send it to a ground station. Around this payload, engineers build a service platform containing the structure, power supply, thermal control, communication system, onboard computer and attitude-control system.

Launch is the first major challenge. The satellite experiences acceleration, vibration and intense acoustic pressure. Its structure must therefore be light but stiff, while sensitive components are mounted and tested so that they do not resonate dangerously. After separation from the launcher, the environment changes completely: there is almost no atmosphere, heat cannot be removed by convection and electronic components are exposed to radiation.

Electrical power usually comes from photovoltaic panels. Batteries supply the spacecraft during eclipse, when Earth blocks the Sun. A power-control unit regulates voltage and distributes energy to the payload, computer, heaters and transmitters. Because available power is limited, mission operations are planned carefully: a high-power transmitter or instrument may only be activated at specific times.

Thermal control is equally important. In sunlight, exposed surfaces may become very hot; in shadow, they may cool rapidly. Engineers use insulation blankets, reflective coatings, heat pipes, radiators and electrical heaters. The objective is not to keep every part at room temperature, but to keep each component within its qualified operating range.

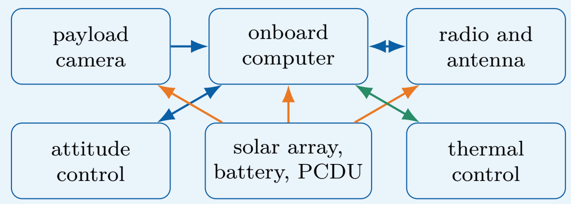
A satellite must also control its orientation, called **attitude**. Sun sensors, star trackers and gyroscopes estimate its orientation. Reaction wheels rotate internally and make the spacecraft turn in the opposite direction. Small thrusters can desaturate the wheels or perform orbital corrections. Since repair is generally impossible, critical functions may be duplicated and the onboard software must detect faults and switch to a safe mode.

Did You Know?

In low Earth orbit, a satellite travels at about $7.8\,\mathrm{km\,s^{-1}}$ and completes one orbit in roughly 90 minutes. It may therefore pass from sunlight to eclipse many times each day.

Engineering Focus

Typical small-satellite architecture

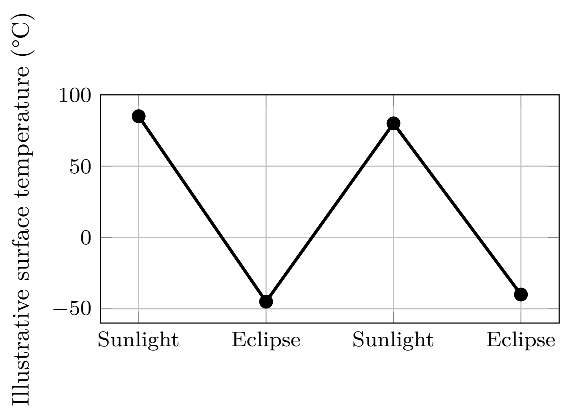


The payload performs the mission, while the platform keeps it powered, pointed, protected and connected. Interfaces between subsystems are as important as the components themselves.

Key Figures

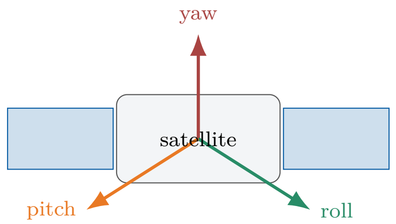
Typical low Earth orbit altitude	400–800 km
Orbital speed	about 7.8 km/s
One orbit	about 90 min
Vacuum pressure	extremely low
Repair after launch	usually impossible

Thermal cycle in orbit



Illustrative values only. Real temperatures depend on orbit, surface coating, orientation, internal dissipation and thermal-control design.

Three-axis attitude control



Sensors estimate orientation; reaction wheels or thrusters create the required rotation around the three axes.

Useful Vocabulary

spacecraft	engin spatial
payload	charge utile
launch load	effort au lancement
solar array	panneau solaire
eclipse	passage dans l'ombre
attitude	orientation
reaction wheel	roue de réaction
star tracker	viseur d'étoiles
thruster	propulseur
redundancy	redondance

Questions

1. Which mechanical loads affect a satellite during launch?
2. Why is convection unavailable in space?
3. What is the role of the battery during eclipse?
4. Name three devices used for attitude determination or control.
5. Why are critical functions sometimes duplicated?

Engineering Challenge

Design a 20 kg Earth-observation satellite. Draw a subsystem diagram and explain how the satellite will generate and store power, control its temperature, point its camera, communicate with Earth and react to one critical failure. Identify two design compromises involving mass, power, cost or reliability.

Speaking Task

Present your satellite in a four-minute mission review. Explain the mission, the architecture and the most serious technical risk. Conclude by defending one design compromise.

Sources

NASA Small Spacecraft Systems Virtual Institute: online resource

European Space Agency – Space Engineering and Technology: online resource

EC10 – Medical Engineering

How can technology improve healthcare?

Learning Objectives

- Describe the measurement chain of a connected medical device.
- Explain the roles of calibration, biocompatibility and risk analysis.
- Design a safe monitoring system combining sensors, software and human supervision.

Engineering News

Wearable sensors, robotic surgery, smart prostheses and remote monitoring are transforming healthcare. These systems can improve diagnosis and quality of life, but they must meet demanding requirements for safety, reliability, privacy and clinical usefulness.

Main Article

Medical engineering combines electronics, mechanics, materials science, computer science and human physiology. A medical device may simply measure a quantity, such as body temperature, or it may act on the patient, as a pacemaker, insulin pump or powered prosthesis does. In every case, the design must begin with a clearly defined medical need and a careful analysis of possible hazards.

Most monitoring devices contain a measurement chain. A sensor converts a physiological quantity into an electrical signal. Because this signal is often weak and noisy, analogue electronics amplify and filter it before an analogue-to-digital converter sends numerical data to a processor. Software then calculates indicators, detects abnormal values and displays information to a patient or healthcare professional. Calibration links the electrical output to a trustworthy physical measurement.

A pulse oximeter, for example, shines red and infrared light through a finger and measures how much light is absorbed. From the changing optical signals, the device estimates blood oxygen saturation and pulse rate. Movement, poor contact or ambient light can disturb the measurement, so the system must detect low-quality signals rather than display a misleading result.

Implantable devices create additional constraints. Their materials must be biocompatible, their energy consumption must be extremely low and their enclosure must protect the electronics from body fluids. A pacemaker continuously monitors cardiac activity and delivers a controlled electrical pulse only when required. This is a closed-loop system: measurement, decision and action are linked.

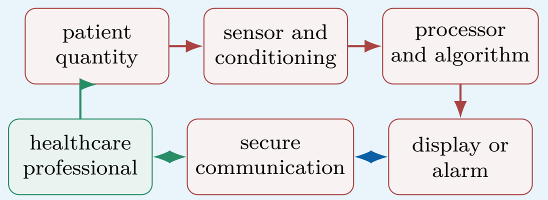
Connected devices can transmit measurements to a smartphone or hospital server. Remote monitoring may allow earlier intervention, but it also introduces cybersecurity and privacy risks. Engineers must protect communication, control software updates and ensure that a network failure cannot place the patient in danger. Clinical validation remains essential: a technically accurate sensor is only useful if its information supports an appropriate medical decision.

Did You Know?

A device can be precise without being accurate. Repeated measurements may be very similar to one another but all shifted away from the true value. Calibration is used to identify and correct this systematic error.

Engineering Focus

Connected patient-monitoring chain

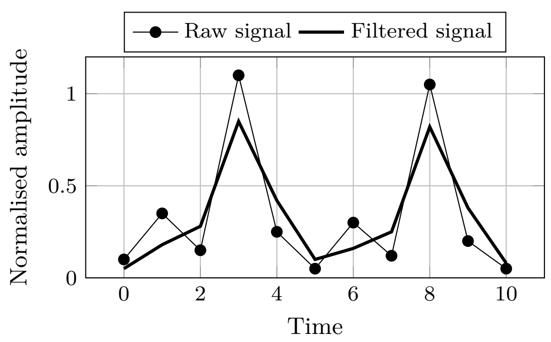


The device provides information, but the complete system includes the patient, the user interface, communication network and medical decision. Each interface must be validated.

Key Figures

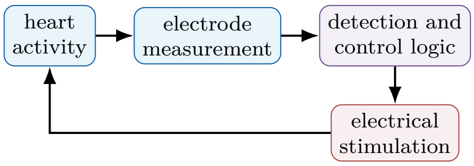
Sensor output	often millivolts or microvolts
Implant operating life	several years
Acceptable false alarm rate	application-dependent
Software traceability	required throughout development
Patient data	confidential

Filtering a physiological signal



Illustrative data. Filtering reduces rapid variations, but excessive filtering may also delay or distort clinically important information.

Closed-loop pacemaker principle



The controller stimulates the heart only when the measured rhythm does not satisfy the programmed criteria. A safe mode must remain available if a sensor or software fault is detected.

Useful Vocabulary

medical device	dispositif médical
implant	implant
prosthesis	prothèse
biocompatible	biocompatible
calibration	étalonnage
signal conditioning	conditionnement du signal
false alarm	fausse alarme
clinical validation	validation clinique
privacy	confidentialité
closed-loop control	commande en boucle fermée

Questions

1. Why must a weak sensor signal be amplified and filtered?
2. Distinguish precision from accuracy.
3. Give two possible disturbances affecting a pulse oximeter.
4. Why must an implant consume very little energy?
5. What makes a pacemaker a closed-loop system?

Engineering Challenge

Design a connected device that monitors an elderly patient at home. Select two physiological or activity measurements, draw the information chain and define alarm conditions. Explain how the system detects poor-quality data, protects privacy, continues operating during a network failure and keeps a healthcare professional in the decision loop.

Speaking Task

Should a connected medical device automatically contact emergency services when it detects a critical event? Present the benefits, risks and safeguards in a four-minute design review.

Sources

World Health Organization – Medical Devices: online resource

U.S. Food and Drug Administration – Medical Devices: online resource

EC11 – Additive Manufacturing

Can 3D printing transform industrial production?

Learning Objectives

- Explain the digital chain from CAD model to finished part.
- Compare additive manufacturing with machining and casting.
- Identify the main design rules, benefits and limitations of metal 3D printing.

Engineering News

Additive manufacturing is now used for rocket engines, aircraft components, medical implants and industrial tooling. Its main advantage is not simply that it prints parts: it allows engineers to manufacture shapes that conventional processes cannot easily produce.

Main Article

Additive manufacturing creates a component layer by layer from a digital model. The process begins with a three-dimensional CAD file. Slicing software divides the geometry into thin horizontal sections and generates the machine instructions. Depending on the technology, a nozzle deposits a molten polymer, a laser melts metal powder, or light solidifies a liquid resin.

This manufacturing route changes the designer's freedom. Internal cooling channels, lattice structures and organic shapes can be integrated into a single component. Topology optimisation can remove material from low-stress zones while preserving load paths. In aerospace engineering, this can reduce mass and therefore fuel consumption. In medicine, a porous implant can be adapted to a patient and encourage bone growth.

The process also has constraints. Overhanging zones may require temporary supports. These supports consume material and must be removed after printing. Thermal gradients can create residual stress, warping or cracks, especially in metal parts. Build orientation affects surface quality, printing time and mechanical properties. The engineer must therefore design both the part and its manufacturing strategy.

A printed component is rarely ready for immediate use. Powder removal, heat treatment, support removal, machining and inspection may be required. Non-destructive testing can detect internal porosity, while dimensional measurement verifies that the geometry is within tolerance. Because properties can depend on layer direction, qualification is essential for safety-critical parts.

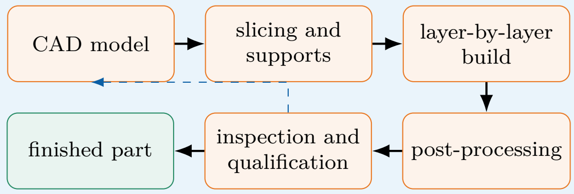
Additive manufacturing is particularly effective for prototypes, customised products, spare parts and low-volume high-value components. It is less competitive for millions of simple identical parts. The best industrial solution is often hybrid: printing creates the complex near-net shape, then conventional machining produces accurate interfaces and smooth functional surfaces.

Did You Know?

A component redesigned for additive manufacturing can combine several conventional parts into one. This reduces assembly operations, fasteners and possible leakage points, but it can make inspection and repair more difficult.

Engineering Focus

Digital manufacturing chain

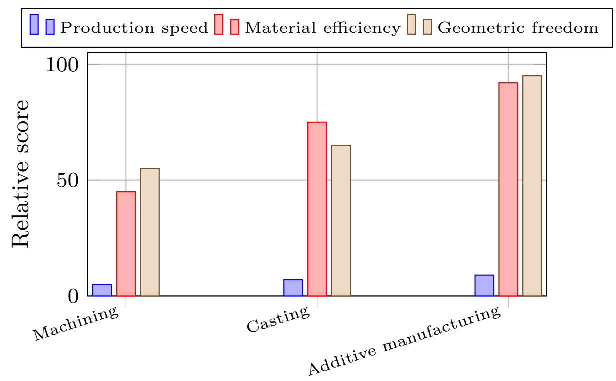


Inspection may lead to a design or parameter change. The chain is therefore iterative rather than strictly linear.

Key Figures

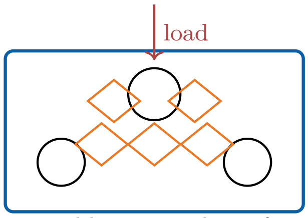
Typical polymer layer	0.10–0.30 mm
Typical metal layer	0.02–0.08 mm
Main input	3D CAD model
Frequent post-processes	heat treatment, machining
Best production range	low volume / high value

Comparing manufacturing routes



Illustrative comparison. The most suitable process depends on quantity, material, tolerance, geometry and total lifecycle cost.

Designing a lightweight bracket



material kept around interfaces and principal load paths

Topology optimisation must be followed by checks on minimum wall thickness, support removal, powder evacuation and machinable reference surfaces.

Useful Vocabulary

additive manufacturing	fabrication additive
slicing	découpage en couches
powder bed	lit de poudre
nozzle	buse
support structure	structure de support
build orientation	orientation de fabrication
warping	déformation / gauchissement
porosity	porosité
post-processing	post-traitement
topology optimisation	optimisation topologique

Questions

1. What information does slicing software generate?
2. Why does build orientation influence a printed part?
3. Give two examples of post-processing operations.
4. Why can mechanical properties be anisotropic?
5. When is additive manufacturing more suitable than mass production?

Engineering Challenge

Redesign a conventional aluminium bracket for metal additive manufacturing. Define the loads and interfaces, propose a lighter geometry, select a build orientation and identify required supports. Then specify two inspection methods and the surfaces that must be machined after printing.

Speaking Task

Will additive manufacturing replace conventional factories? Give a balanced three-minute answer using examples of products for which it is suitable and unsuitable.

Sources

NIST – Additive Manufacturing: online resource
NASA – Additive Manufacturing: online resource

EC12 – Smart Materials

Can materials react to their environment?

Learning Objectives

- Identify the stimulus and response of several smart materials.
- Explain how shape-memory and piezoelectric effects are used in engineering systems.
- Evaluate performance, fatigue, control and lifecycle constraints.

Engineering News

Smart materials can sense or respond to temperature, stress, electricity, light or magnetic fields. They are used in medical devices, vibration control, precision positioning and energy-efficient buildings, where one compact material can perform both structural and functional roles.

Main Article

A conventional material is selected mainly for passive properties such as stiffness, strength or thermal conductivity. A smart material has a controlled response to an external stimulus. The response may be a change in shape, electrical charge, colour, viscosity or magnetic behaviour. Engineers use this coupling to create sensors, actuators and adaptive structures.

Shape-memory alloys are a well-known example. Nickel-titanium can be deformed in a low-temperature phase and recover a previously defined shape when heated. This effect results from a reversible change in crystal structure. A thin wire can therefore act as a compact actuator: electrical current heats it, the wire contracts, and a spring or external load returns it when it cools. Applications include medical stents, valves and lightweight mechanisms.

Piezoelectric materials couple mechanical and electrical energy. When compressed or bent, they produce electrical charge; this is the direct effect used in force, pressure and vibration sensors. When voltage is applied, they deform slightly; this inverse effect is used in ultrasound transducers, fuel injectors and nanometre-scale positioning systems. Their movement is small, but it is rapid and precise.

Other smart materials provide different functions. Electrochromic layers change optical transmission when a voltage is applied, allowing windows to control solar gain. Magnetorheological fluids become more resistant to flow in a magnetic field and can adjust the damping of a suspension. Self-healing polymers may release a repair agent when a crack ruptures embedded microcapsules.

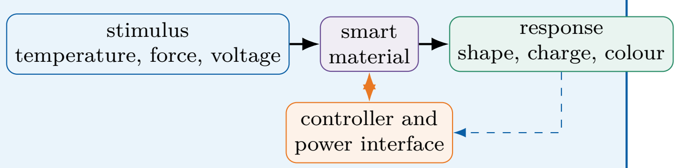
These materials do not act intelligently by themselves. A useful engineering system still requires sensors, power electronics, control logic and safe limits. Performance can change with temperature, fatigue and ageing. The designer must also consider response time, energy consumption, recyclability and what happens when the active effect fails. A passive safe state is often essential.

Did You Know?

A piezoelectric element can be used as both a sensor and an actuator. The same physical coupling converts mechanical energy into electrical energy and electrical energy into mechanical motion.

Engineering Focus

Stimulus–material–response chain

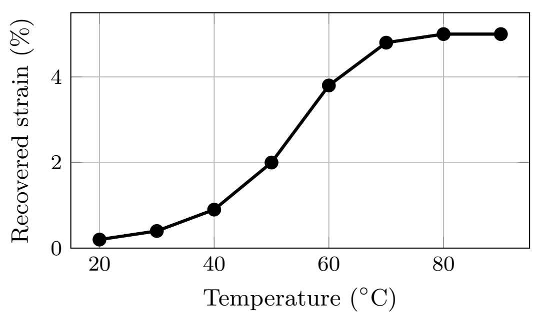


Closed-loop control measures the actual response and adjusts the stimulus. This improves accuracy and compensates for temperature variation or ageing.

Key Figures

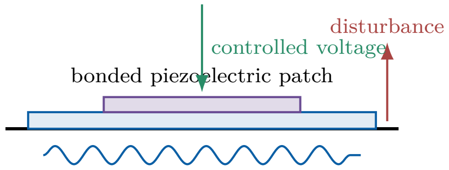
Shape-memory alloy response	several percent strain
Piezoelectric actuator motion	micrometres
Piezoelectric response	very fast
Electrochromic transition	seconds to minutes
Critical design issue	fatigue and ageing

Shape recovery with temperature



Illustrative response of a shape-memory actuator. The transformation occurs over a temperature range rather than at one exact temperature.

Piezoelectric vibration control



structural vibration measured and reduced by an opposing actuator force

A sensor measures vibration, the controller calculates a corrective command and the piezoelectric patch produces a small deformation that opposes the motion.

Useful Vocabulary

smart material	matériau intelligent
stimulus	sollicitation / stimulus
shape-memory alloy	alliage à mémoire de forme
phase transformation	transformation de phase
piezoelectric effect	effet piézoélectrique
strain	déformation relative
damping	amortissement
self-healing	autoréparant
ageing	vieillissement
fail-safe state	état sûr en cas de panne

Questions

1. What distinguishes a smart material from a passive material?
2. How can a shape-memory wire act as an actuator?
3. Explain the direct and inverse piezoelectric effects.
4. Why is closed-loop control useful with smart materials?
5. Give two limitations that must be considered during design.

Engineering Challenge

Design an adaptive bicycle helmet ventilation system using a smart material. Define the stimulus, material response and mechanical function. Add a simple control strategy, estimate the required movement and explain how the helmet remains safe if the active material or power supply fails.

Speaking Task

Choose one smart material and present an everyday product that could use it. Explain the benefit, technical limitation and environmental impact in a three-minute product pitch.

Sources

NIST – Materials Research: online resource

European Space Agency – Space Engineering and Technology: online resource

EC13 – Smart Cities

Can connected systems make cities more sustainable?

Learning Objectives

- Describe the architecture of a connected urban service.
- Explain how sensing, communication and control can reduce resource consumption.
- Discuss interoperability, privacy, resilience and social inclusion.

Engineering News

Cities are installing connected lighting, traffic and environmental-monitoring systems. The most successful projects do not simply collect data: they convert measurements into useful decisions while protecting citizens and keeping essential services available during failures.

Main Article

A smart city uses sensors, communication networks and software to improve urban services. A typical system begins with distributed devices that measure traffic flow, air quality, noise, temperature, water level or electricity consumption. Data are transmitted to a local controller or a digital platform, where algorithms detect trends and select an action.

Street lighting provides a simple example. Conventional lamps operate at full power for most of the night. An adaptive system can combine a light sensor, a motion detector and a timetable. The lamps dim when streets are empty and return to a safe level when a pedestrian or vehicle is detected. Energy savings are possible, but minimum illumination, reaction time and failure behaviour must be specified carefully.

Traffic management is more complex. Cameras, induction loops and connected vehicles estimate queue lengths. A controller can modify signal timing to reduce waiting time and emissions. The local optimisation of one junction, however, may create congestion elsewhere. Engineers therefore model the complete network and define priorities for buses, emergency vehicles, cyclists and pedestrians.

Smart buildings and smart grids can also cooperate. Occupancy sensors adjust heating and ventilation, while a district energy manager coordinates photovoltaic generation, batteries and electric-vehicle charging. This requires devices from different manufacturers to exchange data. Interoperability and open communication standards are therefore essential.

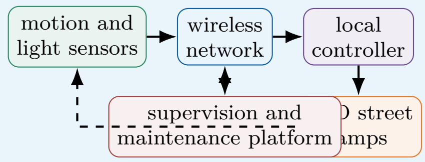
Connectivity also creates risks. Location or camera data may reveal personal behaviour. A network failure must not stop traffic lights, water pumps or emergency systems. Good design separates critical functions, stores only necessary data and includes local fallback modes. Finally, a city is not truly smart if services are inaccessible to people without smartphones or digital skills.

Did You Know?

A digital twin is a computer model linked to measurements from the real city. Engineers can use it to compare traffic, energy or flood-management scenarios before changing physical infrastructure.

Engineering Focus

Architecture of an adaptive lighting service

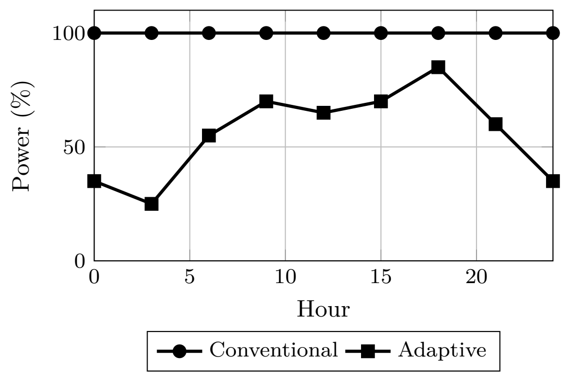


The essential control loop is local: sensing, decision and lighting remain available even if the central platform is disconnected.

Key Figures

Connected devices	thousands per district
Useful lamp response time	a few seconds
Important design property	interoperability
Critical service requirement	fallback mode
Main ethical concern	privacy

Lighting power profile



Illustrative daily profiles. Adaptive control reduces power while maintaining higher levels during periods of activity.

Useful Vocabulary

sensor network	réseau de capteurs
traffic flow	flux de circulation
street lighting	éclairage public
occupancy	occupation / présence
interoperability	interopérabilité
digital twin	jumeau numérique
fallback mode	mode de repli
outage	panne / interruption
privacy	vie privée
urban planning	urbanisme

Questions

1. Describe the data path in a smart-city system.
2. How can adaptive lighting save energy without reducing safety?
3. Why can local traffic optimisation be insufficient?
4. What does interoperability mean in this context?
5. Give two ways to make a connected urban service resilient.

Engineering Challenge

Design a smart energy-management system for a school district. Select sensors and controlled equipment, draw the information flow, define one energy-saving rule and describe the safe fallback mode. Explain how personal data will be minimised.

Speaking Task

Should a city use cameras and connected sensors to reduce congestion? Present one technical benefit, one privacy risk and one condition for public acceptance.

Sources

European Commission – Smart Cities: online resource
NIST – Smart Connected Systems: online resource

EC14 – Water Treatment

How can engineering provide safe drinking water?

Learning Objectives

- Identify the principal stages of drinking-water treatment.
- Explain filtration, disinfection and reverse osmosis.
- Select a robust treatment chain from water-quality and local constraints.

Engineering News

Membrane filtration, ultraviolet disinfection and low-energy desalination are improving access to safe water. Yet the best solution is not always the most advanced one: reliability, operator skills, spare parts and energy availability often determine whether a plant succeeds.

Main Article

Raw water may contain suspended particles, microorganisms, organic matter, dissolved salts and chemical pollutants. No single process removes every contaminant, so drinking-water plants combine several stages. The treatment chain is selected from the quality of the source and the required quality of the final water.

Screens first remove leaves and large debris. During coagulation, a chemical reagent destabilises very small particles that would otherwise remain in suspension. Gentle mixing then creates larger flocs. In a sedimentation tank, these flocs settle under gravity and form sludge that must be collected and treated.

Filtration removes the particles that remain. Sand filters trap material inside a porous bed, whereas activated carbon also adsorbs some organic molecules responsible for taste, odour or pollution. Membrane systems use pores of controlled size. As the membrane becomes dirty, pressure loss increases, so periodic backwashing or chemical cleaning is necessary.

Disinfection inactivates harmful microorganisms. Chlorine can maintain protection throughout the distribution network, while ultraviolet light acts rapidly without leaving a disinfectant residual. Engineers often combine methods and monitor turbidity, pH, conductivity and disinfectant concentration continuously.

Reverse osmosis is used for desalination. Pumps apply a pressure greater than the natural osmotic pressure, forcing water through a semi-permeable membrane while most salts remain in a concentrated brine. The process produces high-quality water but requires energy, pretreatment and careful brine management.

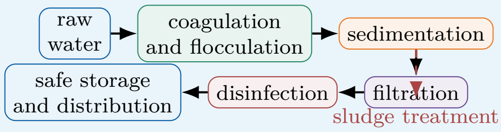
For a remote community, maintainability may be more important than maximum performance. Gravity flow, modular filters, solar pumping and simple sensors can reduce operating costs. A robust system must also include safe storage and prevent treated water from being contaminated again.

Did You Know?

Clear-looking water is not necessarily safe. Pathogenic microorganisms are invisible, so treatment performance must be verified by measurements and microbiological analysis.

Engineering Focus

Multi-barrier treatment chain

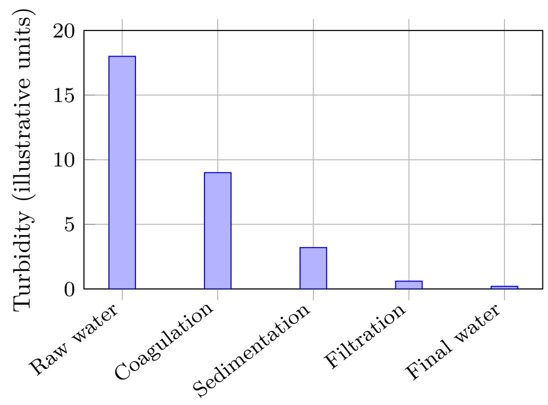


Each stage is a barrier. If one process performs less effectively, the remaining barriers reduce the risk of unsafe water reaching consumers.

Key Figures

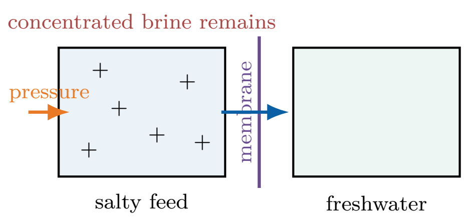
Typical raw-water turbidity	highly variable
Filtered-water target	very low turbidity
Reverse-osmosis driving force	high pressure
Main maintenance issue	fouling
Essential final step	safe storage

Turbidity through the treatment chain



Successive barriers progressively reduce suspended matter. Actual values depend on the source and plant operating conditions.

Reverse osmosis principle



Useful Vocabulary

raw water	eau brute
coagulation	coagulation
floc	floc
sedimentation	décantation
activated carbon	charbon actif
backwashing	lavage à contre-courant
disinfection	désinfection
turbidity	turbidité
reverse osmosis	osmose inverse
brine	saumure

Questions

1. Why does a plant use several treatment stages?
2. Explain the difference between coagulation and sedimentation.
3. What additional function can activated carbon provide?
4. Compare chlorine and ultraviolet disinfection.
5. Why are pretreatment and maintenance important in reverse osmosis?

Engineering Challenge

Design a compact treatment system for a village beside a turbid river. Choose the treatment stages, energy source and monitoring sensors. Explain routine maintenance and the action taken when one measurement exceeds its safe limit.

Speaking Task

Compare a central treatment plant with small local treatment units. Which solution is more suitable for a remote region, and under which conditions?

Sources

World Health Organization – Drinking Water: online resource

U.S. Environmental Protection Agency – Drinking Water: online resource

EC15 – Future Transport

How will people and goods move tomorrow?

Learning Objectives

- Compare transport modes using energy, capacity and infrastructure criteria.
- Explain the perception–decision–action chain of an autonomous vehicle.
- Evaluate a multimodal solution rather than a single vehicle technology.

Engineering News

Battery buses, hydrogen trains, autonomous shuttles and electric aircraft are being tested around the world. The largest environmental gains, however, may come from combining efficient vehicles with public transport, active mobility and better use of existing infrastructure.

Main Article

Transport engineering must move people and goods safely, reliably and with limited energy and land use. Because journey length, passenger capacity and infrastructure differ, no propulsion technology is ideal for every application.

Battery-electric vehicles are efficient because electric motors convert a large fraction of stored energy into motion and can recover energy during braking. They are well suited to urban buses, cars and delivery vehicles that return regularly to a charging point. Long-distance freight requires larger batteries, faster charging or another energy carrier. Hydrogen fuel cells may reduce vehicle mass for some heavy applications, but producing, compressing and converting hydrogen introduces additional energy losses.

Rail carries many passengers or large freight loads with low rolling resistance. Electrified trains can use energy directly from the grid and do not need to carry a large energy store. High-speed rail can replace some short-haul flights when stations connect city centres and total door-to-door time is competitive.

Autonomous vehicles add a digital engineering challenge. Cameras provide detailed images, radar measures distance and relative speed, and lidar creates a three-dimensional representation. Sensor fusion combines these signals. Software then identifies objects, predicts their motion, plans a safe trajectory and commands steering and braking. Adverse weather, unusual road layouts and human behaviour make validation difficult.

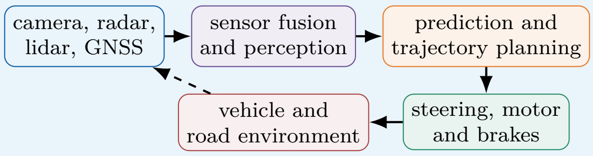
Future mobility is therefore a system problem. A journey may combine walking, bicycle, train and an on-demand shuttle. Ticketing, timetables and physical interchanges must work together. Engineers must also consider accessibility, affordability, charging networks, vehicle manufacture and the carbon intensity of the energy supply. Replacing every conventional car with another type of car would not by itself eliminate congestion or the need for parking space.

Did You Know?

The energy efficiency of a vehicle is only one part of transport efficiency. Occupancy is equally important: a nearly empty heavy vehicle may consume more energy per passenger than a fuller one.

Engineering Focus

Autonomous driving control chain

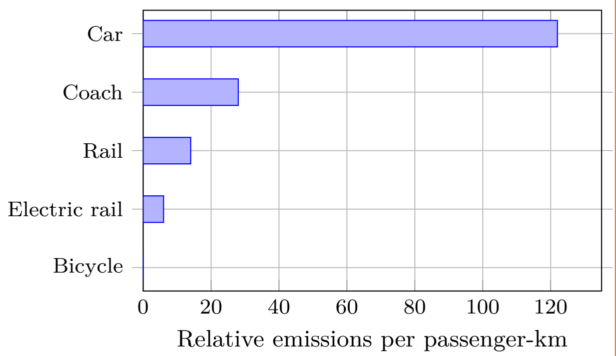


The loop must continue safely even when a sensor is degraded. Redundancy, plausibility checks and a minimum-risk manoeuvre are therefore central design features.

Key Figures

Electric motor efficiency	high
Rail rolling resistance	low
Autonomous control cycle	repeated continuously
Important system metric	passenger-km
Main urban constraint	limited space

Illustrative operational emissions



Illustrative comparison only. Actual emissions depend strongly on occupancy, vehicle, energy source, route and lifecycle boundaries.

Multimodal journey

home → walking / bicycle → rail → shared shuttle → destination

A successful system minimises waiting, transfer distance and uncertainty between modes.

Useful Vocabulary

freight	fret
short-haul flight	vol court-courrier
rolling resistance	résistance au roulement
sensor fusion	fusion de capteurs
trajectory planning	planification de trajectoire
minimum-risk manoeuvre	manœuvre de risque minimal
shared mobility	mobilité partagée
interchange	pôle d'échanges
occupancy rate	taux de remplissage
passenger-kilometre	voyageur-kilomètre

Questions

1. Why is no propulsion technology suitable for every journey?
2. Give one advantage of electrified rail over battery vehicles.
3. Describe the autonomous perception–decision–action loop.
4. Why is vehicle occupancy important when comparing transport modes?
5. Which features make a multimodal journey attractive to users?

Engineering Challenge

Propose a low-carbon transport plan between a suburban school and a city centre. Compare at least three modes using capacity, journey time, energy, accessibility and infrastructure. Include one solution for the first and last kilometre.

Speaking Task

Should city centres restrict private cars? Defend a position using technical, environmental and social arguments, and propose one realistic alternative.

Sources

International Energy Agency – Transport: online resource

European Commission – Mobility and Transport: online resource

EC16 – Cybersecurity

How can engineers protect connected systems?

Learning Objectives

- Distinguish confidentiality, integrity and availability.
- Explain defence in depth for a connected engineering system.
- Analyse technical risk using threats, vulnerabilities and consequences.

Engineering News

Industrial systems, vehicles, hospitals and energy networks increasingly depend on connected software. Cybersecurity is therefore a safety and continuity issue as well as an information-technology issue: a digital attack can produce physical consequences.

Main Article

A connected engineering system contains sensors, controllers, software, communication links and actuators. Each interface can become an entry point for accidental failure or malicious action. Cybersecurity aims to preserve three central properties: confidentiality prevents unauthorised disclosure, integrity prevents unauthorised modification, and availability keeps the service usable.

Risk depends on more than the existence of a threat. Engineers identify valuable assets, possible attackers, vulnerabilities and consequences. A weak password on a non-critical display is not the same risk as an unprotected remote command to a water-treatment pump. Safety analysis and cyber-risk analysis must therefore be coordinated.

No single protection is sufficient. Defence in depth combines several independent layers. Devices should authenticate users and other devices. Communications can be encrypted, software should be updated, and networks should be segmented so that an office computer cannot directly control a critical machine. Access rights follow the principle of least privilege: each account receives only the permissions required for its task.

Monitoring is also necessary. Logs, alarms and anomaly detection can reveal unexpected traffic or commands. When an incident occurs, operators need a prepared response: isolate affected equipment, maintain a safe physical state, restore trusted software and verify operation. Offline backups help recovery, but they must be tested rather than merely stored.

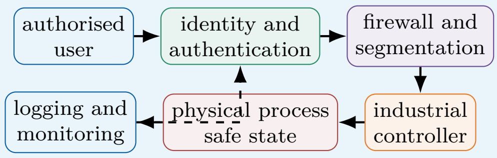
Security creates engineering trade-offs. Encryption and authentication use processing time and energy. Updates may interrupt production, while legacy equipment may not support modern protocols. A secure design therefore begins with requirements and threat modelling, not with a protective tool added at the end. Human factors matter too: clear procedures and training reduce errors and social engineering attacks.

Did You Know?

Safety and cybersecurity can conflict. An emergency door should open quickly for evacuation, yet access control should prevent intrusion. The final design must satisfy both requirements.

Engineering Focus

Defence-in-depth architecture

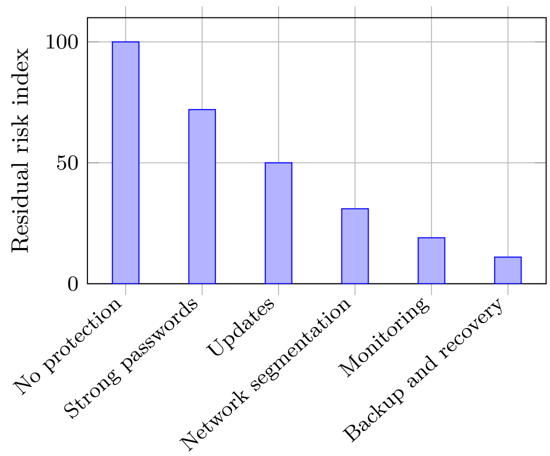


If one layer fails, another should still reduce the probability or consequence of compromise.

Key Figures

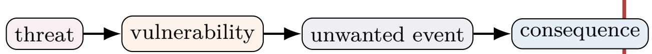
Core security objectives	3
Preferred access principle	least privilege
Critical architecture measure	segmentation
Recovery requirement	tested backup
Essential lifecycle activity	updates

Effect of layered safeguards



Illustrative risk reduction: safeguards are cumulative, but risk never becomes exactly zero and each layer must be maintained.

Risk reasoning



Safeguards may reduce the likelihood of the event, limit its consequences, or improve detection and recovery.

Useful Vocabulary

confidentiality	confidentialité
integrity	intégrité
availability	disponibilité
threat	menace
vulnerability	vulnérabilité
authentication	authentification
encryption	chiffrement
network segmentation	segmentation du réseau
least privilege	moindre privilège
backup	sauvegarde

Questions

1. Define confidentiality, integrity and availability.
2. Why should cyber risk be linked to physical consequences?
3. Give four layers of a defence-in-depth strategy.
4. What is the purpose of network segmentation?
5. Why must backups and incident procedures be tested?

Engineering Challenge

Secure a connected greenhouse containing temperature sensors, an irrigation pump and remote access. Identify assets, one threat and one vulnerability. Propose authentication, segmentation, monitoring and a safe local mode if the network is unavailable.

Speaking Task

Is a fully connected system always better than an isolated one? Compare functionality, maintenance, security and resilience using one engineering example.

Sources

NIST – Cybersecurity Framework: online resource

European Union Agency for Cybersecurity: online resource

EC17 – Internet of Things

Connecting billions of devices

Engineering News

Current developments illustrate the rapid evolution of this engineering field and its impact on society.

Main Article

Internet of Things is an important domain of modern engineering. Engineers combine scientific knowledge, modelling, experimentation and digital technologies to design systems that are safer, more efficient and more sustainable.
A complete engineering system integrates sensors, actuators, electronics, software and communication. Design choices always involve compromises between performance, cost, reliability, safety and environmental impact.
Engineers analyse requirements, model the system, build prototypes, test performance and continuously improve the final solution. International standards and risk analysis play a central role throughout the design process.
Future innovations will rely on multidisciplinary teams capable of combining mechanics, electronics, computer science, materials engineering and artificial intelligence.

Internet illustration for Internet of Things

Engineering Focus

Requirements; modelling; simulation; optimisation; validation; reliability; sustainability; innovation.

Useful Vocabulary

system	système
sensor	capteur
actuator	actionneur
software	logiciel
reliability	fiabilité
safety	sécurité
maintenance	maintenance
innovation	innovation

Questions

1. Explain the engineering problem addressed in this chapter.
2. Which disciplines are involved?
3. Why is modelling useful?
4. Why is testing essential?
5. What are the main technical challenges?
6. Which environmental impacts should engineers consider?
7. Which future innovation seems the most promising?

Engineering Challenge

Propose an engineering solution related to this topic and justify your technical choices.

Sources

IEEE: online resource

EC18 – Drones

Engineering autonomous aerial systems

Engineering News

Current developments illustrate the rapid evolution of this engineering field and its impact on society.

Main Article

Drones is an important domain of modern engineering. Engineers combine scientific knowledge, modelling, experimentation and digital technologies to design systems that are safer, more efficient and more sustainable.
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Internet illustration for Drones

Engineering Focus

Requirements; modelling; simulation; optimisation; validation; reliability; sustainability; innovation.

Useful Vocabulary

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Engineering Challenge

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Sources

IEEE: online resource

EC19 – Biomimetics

Learning from nature

Engineering News

Current developments illustrate the rapid evolution of this engineering field and its impact on society.

Main Article

Biomimetics is an important domain of modern engineering. Engineers combine scientific knowledge, modelling, experimentation and digital technologies to design systems that are safer, more efficient and more sustainable.

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Internet illustration for Biomimetics

Engineering Focus

Requirements; modelling; simulation; optimisation; validation; reliability; sustainability; innovation.

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Engineering Challenge

Propose an engineering solution related to this topic and justify your technical choices.

Sources

IEEE: online resource

EC20 – Circular Economy

Designing products for sustainability

Engineering News

Current developments illustrate the rapid evolution of this engineering field and its impact on society.

Main Article

Circular Economy is an important domain of modern engineering. Engineers combine scientific knowledge, modelling, experimentation and digital technologies to design systems that are safer, more efficient and more sustainable.

A complete engineering system integrates sensors, actuators, electronics, software and communication. Design choices always involve compromises between performance, cost, reliability, safety and environmental impact.

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Internet illustration for Circular Economy

Engineering Focus

Requirements; modelling; simulation; optimisation; validation; reliability; sustainability; innovation.

Useful Vocabulary

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Engineering Challenge

Propose an engineering solution related to this topic and justify your technical choices.

Sources

IEEE: online resource

EC21 – Digital Twin

Creating virtual replicas of physical systems

Engineering News

Recent advances illustrate how Digital Twin is transforming engineering.

Main Article

Digital Twin combines modelling, experimentation and digital technologies to solve complex engineering problems. Engineers analyse requirements, build mathematical models, validate prototypes and optimise performance while considering cost, safety and environmental impact.
Modern systems integrate mechanics, electronics, computer science and communication technologies. Reliable measurements, simulation and testing remain essential before deployment.
Future developments will rely on multidisciplinary engineering teams capable of designing sustainable and resilient technical systems.

Internet illustration: Digital Twin

Engineering Focus

Requirements; modelling; simulation; optimisation; validation; sustainability; reliability; innovation.

Useful Vocabulary

system	système
model	modèle
sensor	capteur
simulation	simulation
performance	performances
reliability	fiabilité
efficiency	rendement
innovation	innovation

Questions

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Engineering Challenge

Propose a realistic engineering solution related to this topic.

Sources

IEEE: online resource

EC22 – Industry 4.0

Connected manufacturing

Engineering News

Recent advances illustrate how Industry 4.0 is transforming engineering.

Main Article

Industry 4.0 combines modelling, experimentation and digital technologies to solve complex engineering problems. Engineers analyse requirements, build mathematical models, validate prototypes and optimise performance while considering cost, safety and environmental impact.
Modern systems integrate mechanics, electronics, computer science and communication technologies. Reliable measurements, simulation and testing remain essential before deployment.
Future developments will rely on multidisciplinary engineering teams capable of designing sustainable and resilient technical systems.

Internet illustration: Industry 4.0

Engineering Focus

Requirements; modelling; simulation; optimisation; validation; sustainability; reliability; innovation.

Useful Vocabulary

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Engineering Challenge

Propose a realistic engineering solution related to this topic.

Sources

IEEE: online resource

EC23 – Autonomous Vehicles

Engineering safe autonomous mobility

Engineering News

Recent advances illustrate how Autonomous Vehicles is transforming engineering.

Main Article

Autonomous Vehicles combines modelling, experimentation and digital technologies to solve complex engineering problems. Engineers analyse requirements, build mathematical models, validate prototypes and optimise performance while considering cost, safety and environmental impact.

Modern systems integrate mechanics, electronics, computer science and communication technologies. Reliable measurements, simulation and testing remain essential before deployment.

Future developments will rely on multidisciplinary engineering teams capable of designing sustainable and resilient technical systems.

Internet illustration: Autonomous Vehicles

Engineering Focus

Requirements; modelling; simulation; optimisation; validation; sustainability; reliability; innovation.

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Engineering Challenge

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Sources

IEEE: online resource

EC24 – Smart Grids

Managing tomorrow's electricity networks

Engineering News

Recent advances illustrate how Smart Grids is transforming engineering.

Main Article

Smart Grids combines modelling, experimentation and digital technologies to solve complex engineering problems. Engineers analyse requirements, build mathematical models, validate prototypes and optimise performance while considering cost, safety and environmental impact.

Modern systems integrate mechanics, electronics, computer science and communication technologies. Reliable measurements, simulation and testing remain essential before deployment.

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Internet illustration: Smart Grids

Engineering Focus

Requirements; modelling; simulation; optimisation; validation; sustainability; reliability; innovation.

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Engineering Challenge

Propose a realistic engineering solution related to this topic.

Sources

IEEE: online resource

EC25 – Marine Renewable Energy

Power from oceans

Engineering News

Recent advances illustrate how Marine Renewable Energy is transforming engineering.

Main Article

Marine Renewable Energy combines modelling, experimentation and digital technologies to solve complex engineering problems. Engineers analyse requirements, build mathematical models, validate prototypes and optimise performance while considering cost, safety and environmental impact.

Modern systems integrate mechanics, electronics, computer science and communication technologies. Reliable measurements, simulation and testing remain essential before deployment.

Future developments will rely on multidisciplinary engineering teams capable of designing sustainable and resilient technical systems.

Internet illustration: Marine Renewable Energy

Engineering Focus

Requirements; modelling; simulation; optimisation; validation; sustainability; reliability; innovation.

Useful Vocabulary

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Engineering Challenge

Propose a realistic engineering solution related to this topic.

Sources

IEEE: online resource

EC26 – Biomedical Imaging

Seeing inside the human body

Engineering News

Recent advances illustrate how Biomedical Imaging is transforming engineering.

Main Article

Biomedical Imaging combines modelling, experimentation and digital technologies to solve complex engineering problems. Engineers analyse requirements, build mathematical models, validate prototypes and optimise performance while considering cost, safety and environmental impact.

Modern systems integrate mechanics, electronics, computer science and communication technologies. Reliable measurements, simulation and testing remain essential before deployment.

Future developments will rely on multidisciplinary engineering teams capable of designing sustainable and resilient technical systems.

Internet illustration: Biomedical Imaging

Engineering Focus

Requirements; modelling; simulation; optimisation; validation; sustainability; reliability; innovation.

Useful Vocabulary

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Engineering Challenge

Propose a realistic engineering solution related to this topic.

Sources

IEEE: online resource

EC27 – Nanotechnology

Engineering at the nanoscale

Engineering News

Recent advances illustrate how Nanotechnology is transforming engineering.

Main Article

Nanotechnology combines modelling, experimentation and digital technologies to solve complex engineering problems. Engineers analyse requirements, build mathematical models, validate prototypes and optimise performance while considering cost, safety and environmental impact.

Modern systems integrate mechanics, electronics, computer science and communication technologies. Reliable measurements, simulation and testing remain essential before deployment.

Future developments will rely on multidisciplinary engineering teams capable of designing sustainable and resilient technical systems.

Internet illustration: Nanotechnology

Engineering Focus

Requirements; modelling; simulation; optimisation; validation; sustainability; reliability; innovation.

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Engineering Challenge

Propose a realistic engineering solution related to this topic.

Sources

IEEE: online resource

EC28 – Quantum Technologies

The next computing revolution

Engineering News

Recent advances illustrate how Quantum Technologies is transforming engineering.

Main Article

Quantum Technologies combines modelling, experimentation and digital technologies to solve complex engineering problems. Engineers analyse requirements, build mathematical models, validate prototypes and optimise performance while considering cost, safety and environmental impact.
Modern systems integrate mechanics, electronics, computer science and communication technologies. Reliable measurements, simulation and testing remain essential before deployment.
Future developments will rely on multidisciplinary engineering teams capable of designing sustainable and resilient technical systems.

Internet illustration: Quantum Technologies

Engineering Focus

Requirements; modelling; simulation; optimisation; validation; sustainability; reliability; innovation.

Useful Vocabulary

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Engineering Challenge

Propose a realistic engineering solution related to this topic.

Sources

IEEE: online resource

EC29 – Sustainable Buildings

Designing low-carbon buildings

Engineering News

Recent advances illustrate how Sustainable Buildings is transforming engineering.

Main Article

Sustainable Buildings combines modelling, experimentation and digital technologies to solve complex engineering problems. Engineers analyse requirements, build mathematical models, validate prototypes and optimise performance while considering cost, safety and environmental impact.

Modern systems integrate mechanics, electronics, computer science and communication technologies. Reliable measurements, simulation and testing remain essential before deployment.

Future developments will rely on multidisciplinary engineering teams capable of designing sustainable and resilient technical systems.

Internet illustration: Sustainable Buildings

Engineering Focus

Requirements; modelling; simulation; optimisation; validation; sustainability; reliability; innovation.

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Engineering Challenge

Propose a realistic engineering solution related to this topic.

Sources

IEEE: online resource

EC30 – High-Speed Transportation

Moving faster with less energy

Engineering News

Recent advances illustrate how High-Speed Transportation is transforming engineering.

Main Article

High-Speed Transportation combines modelling, experimentation and digital technologies to solve complex engineering problems. Engineers analyse requirements, build mathematical models, validate prototypes and optimise performance while considering cost, safety and environmental impact.

Modern systems integrate mechanics, electronics, computer science and communication technologies. Reliable measurements, simulation and testing remain essential before deployment.

Future developments will rely on multidisciplinary engineering teams capable of designing sustainable and resilient technical systems.

Internet illustration: High-Speed Transportation

Engineering Focus

Requirements; modelling; simulation; optimisation; validation; sustainability; reliability; innovation.

Useful Vocabulary

system	système
model	modèle
sensor	capteur
simulation	simulation
performance	performances
reliability	fiabilité
efficiency	rendement
innovation	innovation

Questions

1. What engineering problem is addressed?
2. Which scientific disciplines are involved?
3. Why is modelling important?
4. Which constraints must engineers consider?
5. Which innovation seems the most promising?

Engineering Challenge

Propose a realistic engineering solution related to this topic.

Sources

IEEE: online resource